

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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ABSTRACT. Physical regeneration need not erase what an observer has learned. We prove a causal resource-transfer theorem in which one parameter-independent finite-state program retains its state across every physical regeneration. The actual boundary chain, including its transient states, periodic classes and multiple closed classes, determines the long-run time change. A bounded group inverse is equivalent, on compact boundary families, to cost-universal uniform Cesaro regularity. Together with finite-history product domination and a uniform return-tail certificate, it yields attained robust average risk, finite-model determination, and uniform finite-acquisition and vanishing-discount comparisons at unchanged memory. For a finite model family, increasing the recurrence bound exhausts the unrestricted fixed-memory average value. Complete marked-cycle errors admit a recurrence-sensitive transfer modulus; support-stable controller classes have implementable same-memory upper certificates and all-Borel lower certificates. Actual class-weighted occupation, rather than a pooled ratio or a preparation law, controls predictive compression. A counted controller–predictor lift gives joint control/geometry bounds and matching consequences under stated mass and stability hypotheses. Controlled queue busy periods and noncommuting heralded quantum excursions verify the general theorem independently. Singular holding laws and a task-preserving erasure experiment give sharp memory and tail-sensitive error laws. Uniformity over unrestricted controllers is not inferred from physical mixing, and fixed-exploration geometry is distinguished from optimized control.

1. THE EXPERIMENT AND ITS RETAINED INFORMATION

An observed history fixes the laws of its executable continuations. These laws identify a predictive quotient, but they do not by themselves identify a finite procedure that can maintain that quotient. The procedure must also select actions, transform reports, retain information between acquisitions and produce a task readout. Its occupation law is generated by those operations, not chosen independently of them.

A particularly consequential distinction occurs at a physical regeneration. A new patient, job, specimen or prepared system may be independent of its predecessor conditional on an unknown fixed model. The observer’s finite record need not be independent: it may contain evidence about that model, or a control state that will affect the next experiment. Replacing that record by a prescribed reset label changes the information structure. The correct boundary object is then a finite Markov chain coupled to the durations and rewards of the intervening physical experiments, rather than an iid sequence of complete cycles.

Our main theorem makes this boundary object part of a general experiment-to-risk transfer. It permits arbitrary standard Borel physical state spaces and Borel report rules. It assumes neither a finite-dimensional filter nor a Fisher or covariance geometry. The model parameter is fixed for the entire experiment and is unavailable to the one program being optimized. The raw hypotheses are finite-product response domination and uniform integrability of the physical return time. The

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additional recurrence quantity is computed from the *actually induced* retained-state transition matrix. It does not count as an extra store of information.

The central mechanism has three parts. First, finite-response continuity is proved for a report row that is reused on distinct acquisition coordinates. Second, a group-inverse bound gives a compact, quantitatively regular class of retained programs even when there are several closed classes or a period. Third, a stopped-cycle argument adds physical overshoot to a finite-state Poisson bias. Thus a return moment and a boundary recurrence scale enter separately:

$$(1) \quad |J_N - g| \leq \frac{3\bar{\mu}(2M+1)H}{N} + 2^{1-p}MKN^{-p}, \quad 0 < p \leq 1.$$

The same program has a normalized-discount error bounded by the corresponding expression with N^{-1} and N^{-p} replaced by $1 - \beta$ and $(1 - \beta)^p$. Neither a controller reset nor a free external phase variable is introduced in these comparisons.

The group inverse and its Poisson interpretation are classical [1]. The finite-model conclusion, once compactness has been established, is a finite-intersection argument rather than a minimax exchange. What is proved here is their joint use for one unknown-model Borel causal program, with retained intercycle state, class-dependent physical time, uniform raw-call comparison and same-memory error certification. The recurrence hypothesis is not hidden: Section 3 identifies precisely the compact boundary families on which cost-universal Cesaro convergence is uniform. It also explains how the unrestricted average value of any finite model family is recovered by exhausting these regular classes.

The geometric theorem has a different, explicitly stated role. For a specified exploration, actual class-weighted occupation gives a checkpoint identity and an online lower bound. A stationary, counted controller–predictor lift gives the upper bound. Taking infima yields a joint-control sandwich; additional uniform mass and attainment conditions give a matching optimized memory law. We do not identify an arbitrary fixed-policy quantization rate with an unrestricted control theorem. Likewise, known-law conditional prediction and robust unknown-model control are separate quantifiers unless a common predictive realization has been verified.

Two raw verifications show that the boundary theorem is not a repackaged filtering example. The first is a controlled, partially observed queue with an unbounded workload, where return tails follow from a drift inequality for every legal action. The second uses noncommuting completely positive instruments and heralded termination; the observer retains its classical control state while the physical system is prepared anew. Neither proof is a premise of the general theorem. A worked two-state queue certificate gives the explicit error $17/N$ for an entire feedback class, and a corresponding $17(1 - \beta)$ discount bound. The geometric consequences include nonexpansive propagation, actual rank-changing mass and nonuniform length-biased holding laws.

The comparison with long-run POMDP theory is substantive. Finite-memory approximate optimality in [4], strong uniform values in [5], and average-cost equations in [6] concern broader strategy or model settings but do not give the retained, fixed-cardinality certificate proved here. History-space methods [8] and finite-state average-cost approximation [7] address complementary existence and approximation questions. Controlled restart [9] is not compulsory observer overwrite; intermittent information [10] is not free controller memory. Section 8 gives an assumption-by-assumption comparison, including recent complexity results [11]. We claim neither a general efficient synthesis algorithm nor a new discovery of finite-memory learning [12].

2. A CAUSAL TRANSFER THEOREM WITH RETAINED STATE

2.1. Raw interface and one common program. Let Θ be a nonempty set. A parameter $\theta \in \Theta$ is fixed before the experiment and is not readable by the observer. Physical states, reports and performed audit marks are standard Borel. There is a finite ordinary action set A . A positive

normalized marked kernel specifies the next physical state, the visible report, the visible end-of-cycle flag and the actual audit mark. Costs belong to $[0, 1]$. Audit marks are not feedback unless explicitly included in the visible report. Decisions belong to a compact metric space $\mathcal{D}$, and the bounded marked loss is continuous in the decision, uniformly in the other variables when a quantitative modulus is used.

A visible cycle boundary triggers a prescribed, charged physical preparation. Preparation is call one of the next cycle, and its first ordinary data report is at call two. The physical preparation has a fixed law conditional on θ , independent of previous innovations. The observer label is *not overwritten*. A known preparation-report row, possibly the identity, is part of the program. No extra intercycle seed, tape, counter or exact posterior register is available. All internal randomization is freshly drawn conditional on the stored label. At a scored call the new label is committed before its fixed readout is used; a cleared report workspace is not a second, uncounted readout channel. For simplicity each cycle contains at least one ordinary call; allowing a preparation-only cycle changes no argument.

A stationary M -state program γ consists of action probabilities $\alpha_i(a)$, Borel report rows $k_{ia}(y, j)$, the preparation row, and readouts $v_i \in \mathcal{D}$, where $i, j \in [M]$. These same rows are used at every occurrence, for every θ . Visible report types may distinguish preparation, ordinary and end reports; they are inputs, not uncounted persistent state. Every ordinary action is legal at every ordinary call. Other finite phase interfaces can be encoded in the M labels and are counted there. We write Γ_M for this common program space and start from the fixed label 1.

The physical fresh-start property makes the complete cycle law conditional on an entering label independent of earlier history. For fixed (θ, γ) write

$$(2) \quad P_{ij} = \mathbb{P}_i(I_{\text{next}} = j), \quad d_i = \mathbb{E}_i \tau, \quad r_i = \mathbb{E}_i R, \quad 1 \leq d_i \leq \bar{\mu}, \quad 0 \leq r_i \leq d_i.$$

Here τ includes preparation, every failed acquisition, and the ending call; R is their total bounded loss. The triples $(\tau, R, I_{\text{next}})$ need not be iid across cycles. They form a Markov-renewal kernel on the actual retained labels. Dependence between duration, reward and next label is allowed.

We impose two conditions before optimizing a program.

Finite-response domination. For each action a fix a probability reference σ_a on its visible report space, with visible types kept distinct. For every finite intervention word and every entering label, the report law and its bounded performed-payoff submeasures have L^1 densities relative to the corresponding finite product of these references. The densities are raw experiment data, not chosen for each controller. A preparation coordinate has its own common reference. Zeros and changing support inside a reference are allowed. Domination of single-report marginals alone is not sufficient.

Uniform physical return. There is a displayed decreasing function $a : \mathbb{N}_0 \rightarrow [0, \infty)$ such that

$$(3) \quad \sup_{\theta, \gamma \in \Gamma_M, i} \mathbb{E}_i^{\theta, \gamma}(\tau - T)_+ \leq a(T), \quad a(T) \longrightarrow 0, \quad \bar{\mu} = a(0) < \infty.$$

The supremum can instead range over a specified closed raw-legal architecture; that architecture must then replace Γ_M everywhere. In particular the condition is conditional on *every* entering label. Section 6 derives it from an action-uniform killed-kernel drift, not an independent clock assumption.

Give the ordinary and preparation report rows their weak-star topology in the respective reference L^∞ spaces, action rows their finite simplex topology, and readouts their compact topology. A prescribed preparation row, such as the identity, is a fixed element of that space. Equalities are taken modulo the fixed reference null sets. This is a compact program space.

2.2. The boundary regularity quantity. For any finite stochastic matrix P define

$$(4) \quad \Pi_P = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P^k, \quad G_P = (I - P + \Pi_P)^{-1} - \Pi_P, \quad h(P) = \|G_P\|_\infty.$$

The matrix norm is the maximum absolute row sum. The limit and inverse exist for every finite stochastic matrix, including reducible and periodic ones. This is the group inverse of $I - P$ [1]. Fix $H \geq 0$ and set

$$(5) \quad \Gamma_{M,H}^\Theta = \{\gamma \in \Gamma_M : h(P_\theta^\gamma) \leq H \text{ for every } \theta \in \Theta\}.$$

This is a restriction on the induced recurrence, not an additional memory register. Its nonemptiness is assumed whenever its value is discussed. All Borel controllers satisfying the displayed restriction are included.

For each cycle matrix put

$$(6) \quad D = \text{diag}(d_i), \quad S = I - D^{-1}(I - P), \quad c = D^{-1}r, \quad g_\theta(\gamma) = e_1 \Pi_S c.$$

The matrix S is stochastic because $d_i \geq 1$. If C ranges over the closed classes of P , π_C is its stationary law on C , and w_C is the probability of eventually entering C from label 1, then

$$(7) \quad g_\theta(\gamma) = \sum_C w_C \frac{\pi_C r}{\pi_C d}.$$

The individual ratios are taken *before* mixing classes. In general $(e_1 \Pi_P r)/(e_1 \Pi_P d)$ is a different, incorrect average.

Let C_N be the loss over the first N physical calls, starting at preparation, and set

$$J_N^\theta(\gamma) = \mathbb{E} C_N / N, \quad J_\beta^\theta(\gamma) = (1 - \beta) \mathbb{E} \sum_{t \geq 1} \beta^{t-1} \ell_t.$$

The evaluation index N and the discount factor are not stored in the running machine. Write $V_N^\Theta(M, H)$, $V_\beta^\Theta(M, H)$ and $V_{\text{av}}^\Theta(M, H)$ for the infimum of the respective worst-model losses over the *same* class (5).

Theorem 2.1 (Causal transfer with retained boundary state). *Under finite-response domination and (3), the following hold for every $M \geq 1$ and every nonempty $\Gamma_{M,H}^\Theta$.*

(i) *The class (5) is compact. The average loss exists almost surely for each fixed model and program, has expectation (7), and*

$$(8) \quad V_{\text{av}}^\Theta(M, H) = \min_{\gamma \in \Gamma_{M,H}^\Theta} \sup_{\theta \in \Theta} g_\theta(\gamma) = \sup_{\substack{F \subseteq \Theta \\ 0 < |F| < \infty}} V_{\text{av}}^F(M, H).$$

The class on the right imposes the recurrence bound only for the models in F . A risk level strictly below the left side is excluded by a finite family of joint risk-and-recurrence constraints.

(ii) *Put $B = (2M + 1)H$ and*

$$(9) \quad b_N = \min \left\{ 1, \frac{3\bar{\mu}B + \sum_{s=0}^N a(s)}{N} \right\}.$$

Uniformly over θ and the declared programs,

$$(10) \quad |J_N^\theta - g_\theta| \leq b_N, \quad |J_\beta^\theta - g_\theta| \leq (1 - \beta)^2 \sum_{N \geq 1} N \beta^{N-1} b_N.$$

The same bounds hold between their robust optimal values. Both errors tend to zero. Every cluster point of asymptotically optimal finite-acquisition or discounted programs is average-optimal in (5).

(iii) If $\sup_{\theta, \gamma, i} \mathbb{E}_i \tau^{1+p} \leq K$ for $0 < p \leq 1$, the respective errors in (10) are bounded by

$$(11) \quad \frac{3\bar{\mu}B}{N} + 2^{1-p}MKN^{-p}, \quad 3\bar{\mu}B(1-\beta) + 2^{1-p}MK(1-\beta)^p,$$

capped at one. Boundary recurrence and physical return moments are distinct quantities.

(iv) For a finite model family, let $V_{\text{av}}^\Theta(M)$ be the infimum over all programs in Γ_M , with no recurrence bound. Then

$$(12) \quad V_{\text{av}}^\Theta(M) = \inf_{k \in \mathbb{N}} V_{\text{av}}^\Theta(M, k).$$

Unrestricted attainment is equivalent to the infimum in (12) being reached at some nonempty integer level. No uniform finite- N or discount interchange is asserted after taking this unbounded exhaustion.

(v) For a specified squared-prediction task, the actual class-weighted acquired law and the counted controller–predictor construction of Theorem 5.2 give the joint lower and upper curves (36)–(37). Under the additional uniform mass, global-cover, candidate-stability and attaining-exploration hypotheses of Corollary 5.3, they imply

$$(13) \quad cM^{-2/d} - b_N \leq V_N(M, H) - B_* \leq CM^{-2/d} + b_N,$$

with constants independent of M, N in that declared class. The same-memory numerical root bound is (34); a complete-cycle simulation adds the explicit modulus (19) at the joint simulator recurrence bound. Arbitrary allowed error tolerances do not give lower-risk terms; Theorem 7.2 provides an attained joint memory/calibration/defect law. Thus the geometric part is a conditional matching consequence of the same causal transfer, not a dimension law for every raw kernel.

The theorem includes genuine retained information, but its uniform statement is not over all possible boundary degenerations. The next theorem describes that boundary exactly for cost-universal Cesaro regularity. For an infinite model family an individual program can have $\sup_\theta h(P_\theta) = \infty$; (12) is therefore stated only for finite families.

3. BOUNDARY REGULARITY AND THE PROOF OF THE TRANSFER

3.1. Finite responses and a compact recurrence class.

Lemma 3.1 (A reused row on distinct report coordinates). *Let $f_{n,k}$ be bounded functions on probability spaces (Y_k, σ_k) , with a common bound, converging weak-star to f_k , for $1 \leq k \leq t$. Then for every $u \in L^1(\bigotimes_{k=1}^t \sigma_k)$,*

$$\int u \prod_{k=1}^t f_{n,k}(y_k) d\sigma \longrightarrow \int u \prod_{k=1}^t f_k(y_k) d\sigma.$$

The assertion also holds for nets. In particular, under finite-response domination, every bounded finite-call response is continuous on Γ_M .

Proof. For a rectangle indicator $u = \prod_k 1_{A_k}$ the integral factors into t weak-star pairings. Finite linear combinations of rectangle indicators are dense in L^1 of a finite product of probability spaces. The common product bound controls the approximation error uniformly, which proves the assertion, including for nets. Expand a finite-call program into the finitely many label and action words. Each summand is a raw L^1 response density paired with the product of the selected report rows; action probabilities multiply it continuously. Continuous bounded readout dependence passes by uniform approximation on the compact decision space. Reuse of a row is harmless because its occurrences are evaluated at distinct report coordinates. It would not be harmless on a diagonal report law not dominated by the product reference. $\square$

Theorem 3.2 (An intrinsic criterion for uniform boundary averages). *For a compact family $\mathcal{K}$ of stochastic $M \times M$ matrices, the following are equivalent:*

- (1) $n^{-1} \sum_{k=0}^{n-1} P^k \rightarrow \Pi_P$ uniformly on $\mathcal{K}$;
- (2) $P \mapsto \Pi_P$ is continuous on $\mathcal{K}$;
- (3) $P \mapsto \text{rank}(I - P)$ is locally constant on $\mathcal{K}$;
- (4) $\sup_{P \in \mathcal{K}} h(P) < \infty$.

Moreover, $\{P : h(P) \leq H\}$ is compact and

$$(14) \quad \left\| \frac{1}{n} \sum_{k=0}^{n-1} P^k - \Pi_P \right\|_{\infty} \leq \frac{2h(P)}{n}.$$

This characterizes uniformity for all bounded state rewards and all initial labels, not necessary conditions for a single accidentally constant reward.

Proof. The eigenvalue 1 of a stochastic matrix is semisimple: a nontrivial Jordan block would make its powers unbounded, whereas $\|P^n\|_{\infty} = 1$. Its other unit-circle eigenvalues are also semisimple, and their Cesaro averages vanish. This proves existence of Π_P and the identities

$$(I - P)G_P = G_P(I - P) = I - \Pi_P, \quad \Pi_P G_P = G_P \Pi_P = 0, \quad P \Pi_P = \Pi_P P = \Pi_P.$$

They also prove uniqueness of G_P . Summing the first identity after multiplication by powers of P gives $\sum_{k=0}^{n-1} (P^k - \Pi_P) = (I - P^n)G_P$, hence (14).

Uniform convergence implies continuity of Π_P because each finite Cesaro polynomial is continuous. Its trace is the integer dimension of the eigenspace at 1, so continuity implies local constancy of $\text{rank}(I - P)$. Conversely, near a given matrix of this fixed rank, a small contour around 1 separates that eigenvalue from all others. Constancy of the nullity and semisimplicity imply that the eigenvalues inside the contour are exactly the fixed multiplicity of 1. The contour integral of the resolvent therefore makes Π_P continuous on this relative neighborhood. Then $(I - P + \Pi_P)^{-1} - \Pi_P$ is continuous as well. A finite subcover of $\mathcal{K}$ gives a uniform bound on G_P , which proves all four implications.

Finally suppose $P_n \rightarrow P$ and $\|G_{P_n}\|_{\infty} \leq H$. Along any convergent subsequence of the bounded matrices G_{P_n} , their group-inverse polynomial identities pass to the limit. The limit is the unique group inverse of $I - P$. Thus $h(P) \leq H$. The same argument gives continuity of G and Π on this closed bounded sublevel. Compactness follows from compactness of all stochastic matrices. These finite-dimensional identities are the classical group-inverse mechanism [1]; the cost-universal criterion records exactly which degenerations the uniform causal theorem excludes. $\square$

Lemma 3.3 (Root hitting and the physical time change). *If $h(P) \leq H$, one can choose one label in each closed class such that the expected number of boundary transitions to this set, from any label, is at most $B = (2M + 1)H$. For $1 \leq d_i \leq \bar{\mu}$, the time-changed matrix S in (6) has the same closed classes and class-absorption probabilities as P , and*

$$(15) \quad h(S) \leq 4\bar{\mu}B.$$

If C is a closed class and $g_C = \pi_C r / (\pi_C d)$, the equation

$$(16) \quad (I - P_C)u = r_C - g_C d_C$$

has a solution with $\text{osc}(u) \leq 2\bar{\mu}B$.

Proof. Put the transient block first. Since Π_P has zero transient columns, the transient block of G_P is $(I - Q)^{-1}$. Its row sums are expected times to the recurrent set and are at most H . In a closed class choose j with stationary mass $\pi_C(j) \geq 1/M$. The mean hitting-time identity

$$\mathbb{E}_i T_j = \frac{(G_{P_C})_{jj} - (G_{P_C})_{ij}}{\pi_C(j)} \leq 2MH$$

follows by applying $(I - P_C)G_{P_C} = I - \Pi_{P_C}$ to its j th column and solving the Dirichlet problem with value zero at j . The recurrent row block of G_P is G_{P_C} , so the bound is inherited. Adding the transient and within-class times proves the root bound.

The matrix S holds at i for a geometric number of steps of mean d_i , then makes a P transition. This interpretation includes self-transitions and shows that class absorption is unchanged and that root hitting has expectation at most $\bar{\mu}B$. On a recurrent class its stationary law is proportional to $\pi_C(i)d_i$, proving (7).

For completeness a root hitting bound T_* implies $h(P) \leq 4T_*$. For $\|f\|_\infty \leq 1$ set $v = f - \Pi_P f$, so $\|v\|_\infty \leq 2$. Stop its accumulated sum on first hitting a chosen root. In a recurrent class $\pi_C v = 0$, so the resulting function, assigned zero at the root, solves $(I - P)u = v$ also at that root. Before the recurrent set it solves the same Dirichlet problem. Its norm is at most $2T_*$. Subtracting $\Pi_P u$ gives the unique normalized solution $G_P f$, of norm at most $4T_*$. Apply this to S . For (16), the forcing is centered under π_C and has norm at most $\bar{\mu}$; the same stopped sum, now inside C , has oscillation at most $2\bar{\mu}B$. $\square$

3.2. From boundary cycles to actual acquisitions.

Proof of Theorem 2.1. By Lemma 3.1, finite-call truncated cycle rewards, lengths and exit probabilities are continuous. Truncation changes expected reward and length by at most $a(T)$. It changes exit probabilities by at most $\mathbb{P}(\tau > T) \leq a(T)/(1)$ for integer $\tau > T$, using a truncation one step earlier if necessary. Thus P, d, r are continuous. The closed-sublevel conclusion of Theorem 3.2 makes (5) compact. Lemma 3.3 bounds G_S uniformly, so Π_S and g_θ are continuous on this class.

For a fixed model and program, the finite boundary chain enters a closed class almost surely. Successive returns to a chosen label in that class are regenerative for the *combined* physical experiment and its retained label, with integrable duration and reward. The usual renewal-reward argument, or its stopped-sum proof, gives the almost-sure limit $\pi_C r / \pi_C d$. The expectation is (7), since all averages are bounded by one. This use of a return label in a proof does not physically erase or constrain the observer at a cycle boundary. The renewal facts used here are standard [2].

We prove the quantitative comparison without a stationarity assumption. First suppose the initial boundary label is in a closed class C . At its cycle boundaries, the increments

$$R_k - g_C \tau_k + u(I_{k+1}) - u(I_k)$$

have conditional mean zero by (16). Let K be the first cycle index whose cumulative length exceeds N . Then $K \leq N + 1$. The event that cycle k is reached is determined before its innovations, so the expected stopped sum of these centered increments is zero, by a finite sum of conditional expectations. Removing the portion after call N costs at most the excess length, because the centered per-call loss lies in $[-1, 1]$. The bias costs at most $\text{osc}(u)$.

Here is the overshoot estimate, with no iid assumption. If s is a renewal epoch not exceeding N , the last cycle contributes $(\tau - (N - s))_+$; condition on the entering label and the entire past at s . At most one boundary occurs at each integer s , since durations are positive integers. Therefore

$$(17) \quad \mathbb{E}(S_K - N) \leq \sum_{s=0}^N \mathbb{P}(s \text{ is a boundary}) \sup_i \mathbb{E}_i(\tau - (N - s))_+ \leq \sum_{t=0}^N a(t).$$

This proves the bound with $2\bar{\mu}B$ in place of $3\bar{\mu}B$ when starting recurrently.

From a transient label, wait in the proof until the first chosen closed-class root. Its expected physical time is at most $\bar{\mu}B$. The discrepancy between its prefix reward and the eventual class gain times that prefix, truncated at N , is at most that time. Conditional on the reached root and its arrival time not exceeding N , the preceding recurrent estimate applies to the remaining horizon; its overshoot bound is at most the displayed full-horizon bound. The eventual class has probabilities w_C . Adding the prefix gives (9). In particular $b_N \rightarrow 0$, because Cesaro averages of $a(t) \rightarrow 0$ vanish.

The Abel comparison uses only bounded per-call losses. The identity $\mathbb{E}\ell_N = NJ_N - (N-1)J_{N-1}$ gives, after multiplying by $(1-\beta)\beta^{N-1}$ and summing,

$$(18) \quad J_\beta = (1-\beta)^2 \sum_{N \geq 1} N\beta^{N-1} J_N.$$

These coefficients are nonnegative and sum to one. Hence (10); the right side tends to zero by splitting off a fixed finite range. Under the moment assumption, fix a model and program and denote its conditional lifetime from label i by τ_i . Instead of interchanging a supremum with a sum, use the finite number of entering labels in (17):

$$\sum_{s=0}^N \max_i \mathbb{E}_i(\tau_i - s)_+ \leq \sum_{i=1}^M \mathbb{E}_i[\tau_i \min\{\tau_i, N+1\}] \leq MK(N+1)^{1-p} \leq 2^{1-p} MKN^{1-p}.$$

The factor M can be removed if one common stochastic dominating lifetime has this moment bound. A conditional moment bound alone does not justify that removal. For a geometric variable G on $\mathbb{N}$ with success probability $1-\beta$, concavity gives $(1-\beta)\mathbb{E}G^{1-p} \leq (1-\beta)^p$. Applying this to (18) proves (11). Taking supremum over models and infimum over the *same* class preserves each uniform bound.

The worst-model average is lower semicontinuous on a compact class, so it attains its minimum. For a proposed level v , each set

$$\{\gamma \in \Gamma_M : h(P_\theta^\gamma) \leq H, g_\theta(\gamma) \leq v\}$$

is closed in the compact ambient program space: any convergent net in it has bounded group inverse, continuous time-change projection and the same inequalities at the limit. Finite intersection now proves (8), including the recurrence constraints in the finite subfamilies. There is no exchange of infimum and supremum. Uniform comparisons and lower semicontinuity give the cluster-point assertion.

Finally, for finitely many models each particular program has a finite maximum of the finitely many numbers $h(P_\theta)$. The union of the integer sublevels is therefore all of Γ_M . Taking infima proves (12). If an unrestricted optimizer exists it lies in some level; conversely a level attaining the unrestricted value has an optimizer by part (i). This proves the attainment equivalence without a claim of uniformity as $H \rightarrow \infty$. Part (v) is proved in Sections 4–7: those arguments use only parts (i)–(iii) for existence and time comparison, establish their own geometric and defect inequalities, and then combine them with (10). There is no use of part (v) in their proofs. $\square$

4. CAUSAL DEFECTS AND SAME-MEMORY IMPLEMENTATION

4.1. A defect of complete conditional cycles. A cycle comparison in this section includes the physical preparation, all performed costs and audits, visible reports and flags, actions, duration, and the *next retained label*, conditional on every entering label. We use $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. A comparison of report marginals alone is insufficient. For the next proposition the two endpoints share a program interface; a simulator reduction to this situation is stated afterwards.

Theorem 4.1 (Recurrence-sensitive causal risk transfer). *Suppose corresponding complete conditional cycle laws have total variation at most ε , uniformly over models, legal programs and entering labels. Suppose both endpoints obey the same (3), have mean lengths at most $\bar{\mu}$, and satisfy $h(P), h(P') \leq H$. Put*

$$(19) \quad b(\varepsilon) = \inf_{T \geq 1} \{T\varepsilon + 2a(T)\}, \quad H_t = 4\bar{\mu}(2M+1)H, \quad \chi(\varepsilon) = \min\{1, 2b(\varepsilon) + 4H_t[\varepsilon + b(\varepsilon)]\}.$$

Then $|g_\theta(\gamma) - g'_\theta(\gamma)| \leq \chi(\varepsilon)$, and the same bound holds for worst-model values optimized over any one common paired class. A uniform decision-loss error ω adds at most ω . Under a common bound

$\mathbb{E}_i \tau^{1+p} \leq K$, one may replace $b(\varepsilon)$ by

$$(20) \quad 2K^{1/(1+p)} \varepsilon^{p/(1+p)}, \quad 0 < p \leq 1.$$

The constants can be reduced using direct bounds for $h(S)$ and $h(S')$ in place of H_t .

Proof. Couple the whole completed marked cycle, including its length and exit label, maximally. Truncating its reward to the first T calls and its length at T gives bounded functionals in $[0, T]$. Consequently

$$\|P - P'\|_\infty \leq 2\varepsilon, \quad \|r - r'\|_\infty, \|d - d'\|_\infty \leq b(\varepsilon).$$

Alternatively on the mismatch event of probability at most ε , the discrepancy of either reward or length is bounded by $\tau + \tau'$. Holder's inequality gives (20). Equality of complete marked paths includes equality of their rewards for a common readout; readout perturbations are handled separately.

Since $d_i, d'_i \geq 1$ and $r'_i \leq d'_i$, direct division yields

$$\|c - c'\|_\infty \leq 2b(\varepsilon), \quad \|S - S'\|_\infty \leq 2\varepsilon + 2b(\varepsilon).$$

For any two stochastic matrices, writing $E = S' - S$ and using the group-inverse identities gives the exact relation

$$(21) \quad \Pi_{S'} - \Pi_S = \Pi_{S'} E G_S + G_{S'} E \Pi_S.$$

Indeed split the left side as $\Pi_{S'}(I - \Pi_S) - (I - \Pi_{S'})\Pi_S$, and substitute $(I - S)G_S = I - \Pi_S$ and $G_{S'}(I - S') = I - \Pi_{S'}$. As Π_S is stochastic, Lemma 3.3 now proves (19). This argument does not require equal recurrent-class partitions at the endpoints. Changing readout loss by at most ω changes the long-run per-call loss by at most ω . Taking supremum and infimum over a paired common class completes the proof. $\square$

Proposition 4.2 (Primitive comparison and typed composition). *For classical complete marked kernels, a per-call operational error at most κ , and a preparation error at most κ_0 , give a conditional completed-cycle error*

$$(22) \quad \varepsilon \leq \min\{1, \kappa_0 + \bar{\mu}\kappa\}.$$

The same assertion holds for finite-dimensional quantum instruments when κ bounds half the integrated trace-norm error of their unnormalized marked outputs on every input state. The original endpoint supplies the mean length in this one-sided estimate.

A parameter-independent causal simulator must specify its action lift, report transformation, private state, preparation convention, clock, and task readout. If a target observer has M labels and its simulator has R labels, direct execution has at most MR joint labels. To apply Theorem 4.1, its joint boundary matrix must have a stated recurrence bound; an observer bound alone does not imply it. Defects of serial complete-cycle simulations add, and private state counts multiply. One-way simulation gives one-way decision transfer; reverse inequalities require a reverse simulator or an independent lower bound.

Proof. Classically couple the preparations and, while matched, use a maximal coupling of the whole next marked transition, including the next hidden state, report, audit and end flag. Summing the per-call mismatch hazard over the original alive calls gives (22). It is not enough to couple reports and ignore a subsequent hidden-state discrepancy.

Quantum mechanically telescope finite marked words by replacing their first $t - 1$ instrument factors by the original factors and their remaining factors by the approximating factors. The unnormalized input to replacement t is therefore the original alive state. Its trace weights the one-call error, and the subsequent trace-preserving completed instrument cannot increase the trace norm on Hermitian differences. Sum these trace masses, then let the truncation increase; the mean

original lifetime is the sum of alive masses. Preparation is a separate first replacement. This fixes the orientation of the hybrid argument and uses no normalized posterior denominator.

For the simulator, the joint observer/simulator label determines the next legal operation, so the product-state count is literal. Coupling serial marked outputs gives the additive defect by the union bound; applying the preceding theorem at the resulting recurrence bound gives the risk comparison. Task and action compatibility exclude reading future audits or choosing a true-parameter-dependent row. In general these conclusions concern *target-call* average risk. For source physical-time average risk one must change the holding times in (7). If every target call costs exactly c source calls with zero loss on the other calls, the scored source-time rate is the target-call rate divided by c . For variable conversion times there is a separate ratio in each closed class; a ratio of globally pooled means is not justified. $\square$

4.2. A support-stable architecture with primitive recurrence bounds. Fix a finite directed graph $\mathcal{G}$ on $[M]$. Its reflexive reachability relation permits holds even when $\mathcal{G}$ has no self-loop. Each nonterminal update is supported on the reflexive reachability relation of $\mathcal{G}$. The preparation row is the identity. At an ending ordinary report, each outgoing edge of $\mathcal{G}$ is assigned probability at least a fixed rational $\zeta > 0$; remaining mass may use the reachability relation. Assume these constraints are feasible and that from every prepared physical state the first ordinary call ends with probability at least $c_0 > 0$, uniformly over models and actions. Closed classes of the induced boundary chain are exactly the terminal strongly connected components of $\mathcal{G}$. The class includes adaptive report rules and random actions; it prescribes no common observer-reset label.

Proposition 4.3 (A graph certificate for retained recurrence). *For $M \geq 2$, every program in this architecture satisfies*

$$(23) \quad P_{ij} \geq c_0 \zeta \quad ((i, j) \in \mathcal{G}), \quad h(P) \leq H_{\mathcal{G}} := \frac{4(M-1)}{(c_0 \zeta)^{M-1}}.$$

The architecture may have several terminal components and periodic components. For $M = 1$, take $H_{\mathcal{G}} = 0$.

Proof. On the event of ending at the first ordinary call, the incoming label is still i and the ending row gives each designated edge mass at least ζ . This proves the lower bound. Other within-cycle transitions stay within graph reachability, so cannot create a new terminal component. Select one root in each terminal component and a directed path of at most $M - 1$ edges from each vertex to its selected terminal root. The chance of hitting some root in the next $M - 1$ boundary steps is at least $(c_0 \zeta)^{M-1}$. Applying this bound successively by the Markov property gives an expected hitting time at most $(M - 1)/(c_0 \zeta)^{M-1}$. The stopped-Poisson construction in Lemma 3.3 gives $h(P)$ at most four times this quantity. Aperiodicity or a common one-step minorizing distribution is not used. $\square$

4.3. Effective two-sided certification without increasing M . The next result distinguishes an effective input specification from a complexity claim. Assume the raw graph architecture above and (3), with computable bounds. Supply finite executable report partitions, refining the visible types and graph support regions, with complete marked-kernel reconstruction error $\kappa_h \rightarrow 0$. Reconstruction retains the exact joint cell/next-physical-state/audit/flag output and redraws only the within-cell report from the reference conditional law. Supply finite model nets with a controller-uniform average-risk modulus $\Omega(s) \rightarrow 0$, a compact decision net of mesh ρ with loss modulus $\omega_D(\rho)$, and certified finite-history integrations and tail truncations. Proposition 4.2 and Theorem 4.1 provide Ω when a uniform primitive model modulus is available.

Theorem 4.4 (Global lower and implementable upper at the same memory). *Let $V_{\mathcal{G}}^{\Theta}(M)$ be the worst-model average value over all Borel programs of the displayed graph architecture. The supplied*

data yield finite intervals $[L, U]$, with $U - L$ arbitrarily small, and an actual M -state table such that

$$(24) \quad \max\{0, L - e_{h,Q,\rho}\} \leq V_{\mathcal{G}}^{\Theta}(M) \leq \min\{1, U + \Omega(s)\},$$

where, using any valid common recurrence bound for the endpoints,

$$(25) \quad e_{h,Q,\rho} = \chi(\bar{\mu}\kappa_h) + \chi(\bar{\mu}(|A| + M)/Q) + \omega_D(\rho).$$

For partition sizes J_a and decision-net size D_ρ , a deliberately crude enumeration bound is

$$(26) \quad (Q + 1)^{M|A|+M^2 \sum_a J_a} D_\rho^M,$$

with harmless extra finite preparation entries if these are variable. The lower endpoint applies to every Borel program in this architecture, not merely a chosen local optimizer. The upper table operates on the original experiment, not a simulator with additional persistent states.

Proof. Average each Borel report row once on each reference cell. By the definition of reconstruction, running the original row on the reconstructed instrument has exactly the same joint marked law as running this cell-averaged row on the original instrument. This equality includes the retained exit label and holds from every entering label. Convex support and floor constraints are preserved by averaging. Theorem 4.1 and Proposition 4.2 therefore bound the original-to-cell loss by the first term of (25), with the original all-program tail certificate inherited by the reconstructed endpoint through this equality.

For a row having m required edges, reserve the integer mass $Q\zeta$ on each required edge, choosing Q divisible by the denominator of ζ . Distribute the remaining $Q(1 - m\zeta)$ integer units by cumulative rounding of the residual probabilities. This preserves all forbidden zeros and required floors, keeps the actual denominator equal to Q , and gives row total variation at most M/Q . Round action probabilities with error at most $|A|/Q$ and readouts on the decision net. Every rounded table stays in the same graph architecture, with the same uniform tail and recurrence bounds. The cycle coupling gives the second error in (25). Thus *every* Borel program is covered by a legal finite table with that common error.

For each finite table and model-net point, truncate physical cycles and integrate the finite words to enclose P, d, r . Their tail errors follow from (3). The graph determines the number k of closed classes. For $L_S = I - S$, the identity

$$(27) \quad \Pi_S = \frac{[z^{k-1}] \operatorname{adj}(zI + L_S)}{[z^k] \det(zI + L_S)}$$

follows from $\Pi_S = \lim_{z \downarrow 0} z(zI + L_S)^{-1}$ and semisimplicity of zero. The directed forest expansion [3] makes the denominator a sum of nonnegative spanning-forest weights. A forest using the graph edges exists, so it is at least $(c_0\zeta/\bar{\mu})^{M-k}$. Hence certified integrations give arbitrarily narrow enclosures for every finite table's g without dividing by an unverified small number. Finite maxima over models and minima over tables give L ; evaluating the table with least certified upper value gives U . Refining intervals makes $U - L$ arbitrarily small. Equation (26) bounds the number of integer-coordinate rows before simplex constraints and is intentionally an overcount.

Finite-table inclusion alone does not give the displayed lower bound. The all-Borel coverage just proved gives $V_{\mathcal{G}}^F \geq L - e_{h,Q,\rho}$ on the finite model net F ; full-family value is no smaller. For the selected finite table, the controller-uniform model modulus gives its full-family risk at most $U + \Omega(s)$. Clipping to $[0, 1]$ proves (24). $\square$

The theorem deliberately uses a support-stable architecture. Rounding an arbitrary member of $\Gamma_{M,H}^{\Theta}$ can create a new transition between closed classes and need not preserve its recurrence bound. An effective theorem for *all* such spectral sublevels would require another robustness certificate; it is not obtained by silently applying the preceding enumeration.

A table with $P_0 = M|A| + M^2 \sum_a J_a$ probability entries uses at most $O(P_0 \log(Q + 1))$ program bits, in addition to report-partition and decision descriptions. It has only $\lceil \log_2 M \rceil$ persistent label bits. A dyadic sampler uses a stated finite number of fresh coin bits per row; if a rational floor is not dyadic, exact rational sampling may use a variable fresh-coin count with finite expectation, or a dyadic smaller certified floor can be used. Cell location, decision evaluation, temporary arithmetic and planning enumeration are separately charged. No bound in (26) is called efficient, and this implementation does not prove optimal lower bounds for those additional resources.

5. ACQUIRED OCCUPATION AND COUNTED CAUSAL GEOMETRY

5.1. Predictive quotient and the task being compressed. A history determines the conditional laws of all its legal future experiments, including available actions, report events and charged stopping tests. Two histories are predictively equivalent if all these quantities agree. On a standard Borel history space, a countable determining family of future tests gives the measurable map

$$h \longmapsto (\mathbb{E}[T_j \mid h])_{j \geq 1} \in [0, 1]^{\mathbb{N}}.$$

Its fibers are precisely that equivalence relation. Its image need not be Borel solely because this map is Borel. Whenever a standard Borel realization is needed below, it is an explicit hypothesis verified by the raw model: the image has a standard Borel structure, a Borel section, and report and update kernels descend measurably. No measurable quotient theorem is inferred from positivity alone.

Lemma 5.1 (Minimal realized prediction). *Suppose a countable test family determines all executable continuations and is closed under the stated one-step report and action continuations. Suppose its evaluation image S is standard Borel and has a Borel section. Then the evaluation map is a measurable predictive quotient. Any statistic that determines all the tests factors to this quotient on its attained range whenever its own evaluation readouts are measurable. A report kernel and update kernel constant on the evaluation fibers descend to S by composition with the section; the descended kernels do not depend on the chosen section.*

Proof. Equality of coordinates is exactly equality of all determining tests and hence all permitted continuations. The quotient map is Borel coordinatewise. If Z determines the tests by Borel functions $f_j(Z)$, its factor is $z \mapsto (f_j(z))_j$, which lies in S on the attained range. The section composes Borel kernels with Borel maps; fiber constancy makes the result section-independent. Including action-availability and visible clock coordinates prevents identification of histories with different legal future interfaces. This proves the stated minimality, not a prior-free sufficient statistic for every unknown-model family. $\square$

The task coordinates in the geometric inequalities may be a closed response factor of the full future-test quotient under a specified execution; they are not asserted to identify histories having different uncompressed report laws. The remainder of this section is under one specified physical law and exploration, or under a common predictive realization separately verified for an entire model family. For a bounded Hilbert-valued performed audit Y_t , let $p_t = \mathbb{E}[Y_t \mid \mathcal{H}_t]$ at a designated scored data call. A zero-score within-cycle report can still update the predictive state. All updates used by the observer are stationary functions of its counted controller state, stored center, and current visible report type. A phase variable needed for this property must be either presently reported or included among the counted labels. The known-law predictive prior at a physical preparation is fixed; an unknown-parameter posterior carried across preparations is *not* discarded under this convention.

Fix a Q -state exploration controller γ . Conditional on each entering controller label i , let ν_i be the expected sum of point masses at the data-call predictive means during that cycle. With the

closed-class quantities of (7), the actual acquired occupation submeasure is

$$(28) \quad \nu^\gamma = \sum_C w_C \frac{\sum_{i \in C} \pi_C(i) \nu_i}{\sum_{i \in C} \pi_C(i) d_i}.$$

Preparation and other zero-score calls remain in the denominators, so ν^γ can have mass below one. Transient occupation has zero average mass under the return and recurrence bounds. Let B^γ be the corresponding average of $\mathbb{E}[\|Y_t - p_t\|^2 \mid \mathcal{H}_t]$. For any finite submeasure ν , write

$$(29) \quad q_J(\nu) = \inf_{c_1, \dots, c_J} \int \min_j \|v - c_j\|^2 \nu(dv).$$

This is defined on the actual task-response space, not on an ambient parameter chart.

Theorem 5.2 (Occupation geometry with retained control). *For the specified controller and task, a checkpoint encoder with one fixed codebook of J responses has excess risk exactly $q_J(\nu^\gamma)$. Every online observer with at most J fixed response values has the same lower bound, even if it retains information across physical regenerations.*

For a counted implementation let ν_q^γ be the part of (28) where the current control label is q . If the exploration label is retained and each of its Q values has J conditional response labels, total storage is QJ and the checkpoint identity becomes

$$(30) \quad R_{\text{cp}}^\gamma(Q, J) - B^\gamma = \sum_{q=1}^Q q_J(\nu_q^\gamma).$$

Every online implementation with this same conditional labeling convention has this lower bound.

Suppose a global J -center cover in every control fiber has radius r_J , including a valid preparation center. Let $U_{q,a,y}$ be a Borel candidate-valid predictive update and f a task readout of Lipschitz constant L_f . For every pair of exact and candidate states used by this recursion suppose

$$(31) \quad d(U_{q,a,y}s, U_{q,a,y}s') \leq L_{q,a,y}d(s, s').$$

At each preparation the predictor coordinate, but not the exploration label, may be initialized to the known prior. Let G_t be the propagated sum of unit errors since this initialization, so $G_t = 1 + L_t G_{t-1}$ on an updated call. Assume its actual class-weighted data occupation satisfies

$$(32) \quad \mathcal{S}_\gamma = \sum_C w_C \frac{\sum_{i \in C} \pi_C(i) \mathbb{E}_i \sum_{t \text{ data}} G_t^2}{\pi_C d} < \infty.$$

There is one stationary QJ -state controller-predictor with

$$(33) \quad R_{\text{on}}^\gamma(Q, J) - B^\gamma \leq L_f^2 r_J^2 \mathcal{S}_\gamma.$$

If the control boundary matrix has $h(P) \leq H$, the joint implementation has $h(P_) \leq H + 2$. With update numerical error at most δ in the state metric per updated call and readout error at most ξ , the square root of the right side of (33) can instead be bounded by*

$$(34) \quad L_f(r_J + \delta) \sqrt{\mathcal{S}_\gamma} + \xi \sqrt{\nu^\gamma(S)}.$$

The approximate states must remain in the same candidate-valid domain.

Proof. Conditional projection gives $\mathbb{E}\|Y_t - c\|^2 = \mathbb{E}\|Y_t - p_t\|^2 + \mathbb{E}\|p_t - c\|^2$. For any random label, the last term is at least the squared distance to the fixed codebook. Conditional cycle summation and class-by-class renewal reward give (28) and the lower bounds. A full-history checkpoint can choose a nearest center, measurably after breaking ties by the smallest index, proving the identities as infima. On compact task support the optimizing centers may be restricted to its compact closed convex hull in finite dimension; in general an arbitrarily close codebook suffices for the equality of

infima. Formula (30) is the same argument with separate fixed codebooks in the control fibers. No codebook is chosen retrospectively for the realized history.

The online machine stores (q, j) . It runs the original exploration using q alone, computes U at the stored center, and chooses its nearest covered center in the new control fiber. The exact state is used only in this proof. From (31), $e_t \leq L_t e_{t-1} + r_J$ and hence $e_t \leq r_J G_t$. Sum its squared readout error under (28) to obtain (33). The same induction with error $r_J + \delta$, followed by the triangle inequality in the actual occupation L^2 space, proves (34). This is a root-error statement; the full excess bound is its square.

It remains to count and verify the retained boundary state. At preparation the predictor is voluntarily initialized from the known physical prior; the exploration label is unchanged, and later actions ignore the predictor label. Thus the next joint boundary law depends on the entering pair only through q . There are stochastic matrices A (forget the predictor) and K (one full cycle from q to the exiting pair) with $P_* = AK$ and $KA = P$. Consequently

$$\Pi_{P_*} = A\Pi_P K, \quad G_{P_*} = I - \Pi_{P_*} + AG_P K.$$

These follow either by finite Cesaro sums and the group-inverse identities, or by multiplying them directly. Their norms give $h(P_*) \leq 2 + h(P)$. There are exactly QJ possible pairs; neither A nor the proof state supplies additional memory. This is a partial predictor initialization permitted by the raw interface, not an imposed reset of the retained control information. $\square$

Corollary 5.3 (Mass, singular strata and optimized control). *If an actual submeasure $\lambda \leq \nu^\gamma$ has mass $w > 0$ and $\lambda(B(x, r)) \leq Cr^d$ for all centers and $0 < r \leq r_0$, then for all sufficiently large J ,*

$$(35) \quad q_J(\nu^\gamma) \geq \frac{w}{2} \left(\frac{w}{2CJ} \right)^{2/d}.$$

A global cover $r_J \leq C_1 J^{-1/d}$ and a uniform bound in (32) give matching conditional checkpoint and online rates for fixed Q whenever the lower submeasure occurs in at least one control fiber. Its mass is never normalized away. Finite strata, atoms and nonuniformly stable paths are allowed.

For a joint decision problem over all retained M -state controllers with recurrence bound H , the general lower bound is

$$(36) \quad V_{\text{joint}}(M, H) \geq \inf_{\gamma \in \Gamma_{M, H}} \{B^\gamma + q_M(\nu^\gamma)\}.$$

For each $QJ \leq M$, every implementable exploration controller with $h(P) \leq H - 2$ and the preceding uniform geometric hypotheses supplies the upper bound

$$(37) \quad V_{\text{joint}}(M, H) \leq \inf_{\gamma \in \Gamma_{Q, H-2}^{\text{geom}}} \{B^\gamma + L_f^2 r_J^2 \mathcal{S}_\gamma\}.$$

Unused labels can be padded as isolated states. If all admissible explorations have $B^\gamma \geq B_$ and a uniform small-ball lower, and one fixed-size Q_* controller attains B_* with uniform cover and stability constants, then for $M \geq Q_*$, $V_{\text{joint}}(M, H) - B_* \asymp M^{-2/d}$, subject to the displayed recurrence margin and with small M absorbed into constants. These attainment and uniform-mass hypotheses are additional conditions, not consequences of a fixed-policy example.*

Proof. The union of J balls of radius $r = (w/(2CJ))^{1/d}$ has λ -mass at most $w/2$. The remaining mass is at distance at least r from every center, proving (35). Apply the upper bound of Theorem 5.2 in each finite fiber; an actual lower fiber suffices for a lower bound on the sum. Every joint observer's actual policy determines B^γ and ν^γ , and its M fixed readouts give (36). The counted product construction proves (37); the group-inverse margin is exactly the bound just proved. The final matching assertion follows with $J = \lfloor M/Q_* \rfloor$ and the stated uniform lower. Nothing in this reasoning replaces the infimum over control policies by a fixed policy without the extra hypotheses. $\square$

Proposition 5.4 (Exact nonuniform holding geometry without observer overwrite). *At each paid preparation draw a fresh vector V from a given compactly supported law π and an integer lifetime $L \geq 1$. A visible load reveals V ; it is scored at that call and at the following $L - 1$ blank holding calls. Preparation has zero score. The target at each scored call is V , and $\mathbb{E}(L + 1) < \infty$. The incoming observer label is arbitrary and is not forcibly overwritten. Then, over all stationary M -state observers,*

$$(38) \quad R_{\text{on,av}}(M) = R_{\text{cp,av}}(M) = q_M(\nu), \quad \nu(dv) = \frac{\mathbb{E}[L \mid V = v]\pi(dv)}{\mathbb{E}(L + 1)}.$$

For a family of preparations the robust value is $\inf_{|C| \leq M} \sup_{\theta} \int d(v, C)^2 \nu_{\theta}(dv)$, with a single codebook. This direct statement does not require common domination of the preparation family. The achieving boundary rows are identical, so $h(P) \leq 2$.

Proof. At every scored call, regardless of the incoming state or earlier cycles, a fixed M -response observer has squared loss at least its pointwise nearest-codeword error. The physical preparation and holding process are independent of its actions, so its actual time average has precisely the length-biased law in (38). At load choose a nearest center and retain its label through blank reports. This is one stationary M -state program: the load/hold/preparation tags are visible, and the preparation has zero score. Old information is not deleted by the interface, but cannot improve the pointwise bound. Its boundary distribution is independent of the incoming label, giving $P = \Pi_P$ and $G_P = I - P$, of norm at most two. The identical pointwise argument with one fixed codebook proves the robust assertion for arbitrary preparations. It is not a proof of general nondominated Borel-controller compactness. $\square$

6. TWO RAW REALIZATIONS AND PRIMITIVE RETURN CERTIFICATES

6.1. Action-uniform drift instead of an independent clock.

Proposition 6.1 (A killed-kernel return certificate). *Let $Q_{\theta,a}$ be the subprobability kernel of surviving one ordinary call, on the alive physical state. If a measurable $V \geq 1$ satisfies*

$$(39) \quad Q_{\theta,a}V \leq \lambda V, \quad 0 < \lambda < 1, \quad \sup_{\theta} \mathbb{E}_{\text{prep}} V \leq B_0,$$

for every legal action and physical state, then every history-adaptive action rule, hence every retained Borel program, satisfies

$$(40) \quad \mathbb{P}(\tau > T) \leq B_0 \lambda^{T-1} \quad (T \geq 1), \quad a(T) = \frac{B_0 \lambda^{T-1}}{1 - \lambda} \quad (T \geq 1),$$

with $a(0) = 1 + B_0/(1 - \lambda)$. If a better mean bound $\bar{\mu}$ is known, replace each displayed tail bound by its minimum with $\bar{\mu}$ and take $a(0) = \bar{\mu}$. All physical return moments have computable bounds. A finite-dimensional quantum analogue holds when a positive matrix $W \geq I$ satisfies $\sum_z \mathcal{I}_{\theta,a,z}^{\text{alive}}(W) \leq \lambda W$ and $\text{tr}(\rho_{\theta} W) \leq B_0$.*

Proof. Conditional on the entire visible and physical past, every chosen action obeys (39). Integrating its randomized choice preserves the inequality. Induction on the alive physical subprobability gives $\mathbb{E}[V(X_t)1_{\{t \text{ alive}\}}] \leq B_0 \lambda^t$, and $V \geq 1$ bounds survival probability. Summing the integer tail from T onward, with preparation counted as one call, gives (40); its bound is deliberately conservative at the first indices. Polynomially weighted geometric tail sums bound each moment. Quantum mechanically apply the dual inequality to the unnormalized alive state and use $W \geq I$ to dominate its trace. The chosen controller's past can affect its next action but cannot invalidate an inequality holding for every action. Termination can depend on state and action and is not assumed independent of the observations. $\square$

6.2. Controlled queue busy periods with retained statistical evidence.

Theorem 6.2 (A partially observed unbounded-workload realization). *Each preparation starts a workload at $W = 1$. On an ordinary call an action $a \in A$ gives*

$$W' = W + 1 \text{ with probability } p_{\theta,a}, \quad W' = W - 1 \text{ otherwise,} \quad 0 < p_{\theta,a} \leq \bar{p} < 1/2.$$

Hitting zero produces a visible end flag and the next charged preparation. The observer is not overwritten. It receives a finite-alphabet noisy workload report; for example, on a nonending call, $\mathbb{P}(Y = 1 \mid W') = (W' + 1)/(W' + 2)$. A performed bounded audit, such as an independent next-arrival indicator with its specified action/model law, may be scored with a continuous compact readout loss but is not feedback.

This raw experiment satisfies finite-response domination and the return hypothesis for every feedback program. Theorem 2.1 therefore applies on every nonempty $\Gamma_{M,H}^\Theta$. The graph architecture of Proposition 4.3 is a primitive nonreset sufficient class, with $c_0 = 1 - \bar{p}$. For a computably compact parameter family, computable finite report and audit probabilities, and a uniform parameter modulus bounded away from singular transition endpoints, it also verifies Theorem 4.4 on that architecture.

In particular let $\bar{p} = 1/4$, $M = 2$, and require both ending-row label probabilities to be at least $1/4$, while leaving actions and nonending rows arbitrary. For all such programs and models,

$$(41) \quad |J_N^\theta - g_\theta| \leq \min\{1, 17/N\}, \quad |J_\beta^\theta - g_\theta| \leq \min\{1, 17(1 - \beta)\}.$$

These constants count preparation and require no workload truncation in the running controller.

Proof. Reports and flags are finite, so a full-support finite reference on each visible alphabet dominates every finite intervention word. Hidden workloads need not be observed, finite or replaced by a finite belief. Finite histories are countable, so the evaluation quotient of any countable determining family is countable and has a measurable section obtained by selecting the first history in a fixed enumeration. Robust control itself uses the raw kernels and does not require that known-law quotient to be a common posterior for unknown θ .

Take $1 < z < (1 - \bar{p})/\bar{p}$ and $V(w) = z^w$. At $w > 1$, the conditional alive expectation divided by $V(w)$ is $p_{\theta,a}z + (1 - p_{\theta,a})/z \leq \bar{p}z + (1 - \bar{p})/z = \lambda < 1$. At $w = 1$ killing removes the downward term, so the same inequality holds. The prepared value is $B_0 = z$. Proposition 6.1 is therefore uniform over actions chosen using arbitrarily complicated past histories. The probability of ending on the first ordinary call is at least $1 - \bar{p}$, verifying the graph hypothesis independently of the observer.

For effective verification, a truncation to T ordinary calls visits workloads at most $T + 1$. Its action/report/audit words have finite, computable sums of products of the primitive probabilities. The omitted duration and reward are bounded by the displayed drift tail. For two parameter values, coupling each up/down transition by one uniform variable bounds the one-call complete hidden-transition error by $\max_a |p_{\theta,a} - p_{\theta',a}|$, plus the separately specified report/audit error. A preparation error, if any, is counted separately. The same coupling and Theorem 4.1 give the model modulus required by Theorem 4.4. Finite report cells are already exact; no belief discretization or report-only approximation is being substituted for the hidden transition.

We give the numerical constants in (41). The constant- $1/4$ biased random walk dominates every controlled busy period pathwise, by coupling the up/down uniforms until the controlled walk is killed. Its busy duration T_b from one has

$$\mathbb{E}T_b = 2, \quad \text{Var}(T_b) = 6, \quad \mathbb{E}(1 + T_b)^2 = 15, \quad \mathbb{E}(1 + T_b) = 3.$$

These follow from the first- and second-moment difference equations for the stopped biased walk; their derivation is given in Appendix A. Thus $\bar{\mu} \leq 3$ and the all-program overshoot sum is bounded by $\mathbb{E}[(1 + T_b)(2 + T_b)]/2 \leq 9$. If $P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$, then $a, b \geq (3/4)(1/4) = 3/16$. The forcing

$f = r - gd$ has oscillation at most 3, since all its coordinates lie in $[-3g, 3(1 - g)]$. Solving the two-state Poisson equation gives bias oscillation at most $3/(a + b) \leq 8$. There is no transient class. The stopped-cycle argument in (17) therefore gives $(8 + 9)/N$. Its Abel mixture gives $17(1 - \beta)$. This is an explicit bound for the entire declared feedback architecture, not evidence from simulated trajectories. $\square$

6.3. Noncommuting heralded quantum excursions.

Theorem 6.3 (A positive-instrument retained-state realization). *Let a finite-dimensional physical system be freshly prepared by ρ_θ at each heralded boundary, while its classical M -state controller is retained. For each action, let a normalized completely positive instrument produce a visible report and an alive/end flag, and any performed but unreported audit. Suppose report instrument densities are measurable with respect to common probability references and an action-uniform positive-matrix drift as in Proposition 6.1 holds. Then the raw experiment verifies Theorem 2.1 on each nonempty recurrence sublevel, including pure preparations, zero report probabilities, rank changes and noncommuting actions.*

For a concrete family on a qubit, fix $0 < s_1 \neq s_2 < 1$ and unitaries U_a which need not commute. Let $|1\rangle, |2\rangle$ be an orthonormal basis and let $|0\rangle$ denote a fixed unit vector in the same two-dimensional space, not a third state. Put

$$(42) \quad K_a = \text{diag}(\sqrt{s_1}, \sqrt{s_2})U_a, \quad J_{a,j} = \sqrt{1 - s_j} |0\rangle\langle j|U_a, \quad j = 1, 2.$$

The K_a report is alive; the $J_{a,j}$ reports are heralded ends. This family has a state-dependent termination mechanism, $W = I$, $\lambda = \max_j s_j < 1$, and first-call ending probability at least $c_0 = 1 - \lambda$. Any feasible graph architecture from Proposition 4.3 therefore gives a primitive recurrence certificate without observer overwrite.

Computably compact parameter families with a strict common $\lambda < 1$, computable matrix entries, certified integrations on report cells and a complete-instrument modulus verify Theorem 4.4. A bound on a chosen Kraus representation is sufficient input, but the resulting operational instrument norm does not depend on the representation.

Proof. Compose unnormalized instrument densities along a finite action/report word and take its trace. This is its nonnegative finite-word density, whose integral is at most one before completed flags are summed. Weighting the same word by an actually performed bounded audit gives an L^1 payoff density. The construction is valid at zero-probability reports without a Bayes denominator. The raw instrument and physical preparation make the complete conditional cycle depend only on the entering retained label and the fixed parameter. The drift proposition gives the required all-program tail. This proves the hypotheses of Theorem 2.1 directly, without invoking a quantum risk formula.

For (42), $K_a^* K_a + \sum_j J_{a,j}^* J_{a,j} = I$. The alive dual maps I to $U_a^* \text{diag}(s_1, s_2)U_a \leq \lambda I$. Unequal s_j make the ending probability depend on the input state; a jump output is rank one. Distinct noncommuting unitaries change the subsequent instruments, not just a common classical basis. Any extra audit channel used by the task is included in the complete instrument, with its average back-action, and its drift must be checked after that composition. A trace-preserving audit following the alive branch preserves the $W = I$ alive trace inequality.

If A, B are corresponding Kraus factors and $X \geq 0$, then

$$\|AXA^* - BXB^*\|_1 \leq (\|A\| + \|B\|)\|A - B\| \text{tr } X.$$

Sum this estimate over all marked factors and integrate the report reference. It supplies an operational one-call error for Proposition 4.2. Cell averaging must average the *whole completely positive instrument*, not average Kraus matrices and resquare them. Finite-word integrals are certified matrix products, and the drift tail controls the remainder. These statements verify

precisely the effective inputs specified in Theorem 4.4. The observer remains classical with M counted labels; the physical quantum system is not supplied to it as an additional writable memory. Nothing here concerns infinite-dimensional unbounded operators. $\square$

Corollary 6.4 (Raw acquired geometry and genuine singular components). *The hypotheses of Theorem 5.2 have the following independent hidden and quantum verifications for a known preparation and a counted, fixed-size exploration controller whose retained boundary chain has a recurrence certificate.*

For a hidden state with $d + 1$ values and positive preparation π , take first-report likelihoods on the uniform cube $[-1, 1]^d$,

$$(43) \quad g_i(y) = 1 + \eta y_i \quad (i \geq 1), \quad g_0(y) = 1 - \eta \sum_{i=1}^d y_i, \quad 0 < \eta < 1/(2d).$$

After that load, publicly identified stochastic transitions, including permutations, propagate the indicator-audit predictive mean. For a quantum system of dimension q , put $d = q^2 - 1$, choose a traceless Hermitian basis H_j , and use a sufficiently small symmetric cube on which $E(y) = I + \sum_j y_j H_j > 0$. Prepare I/q . With actual probability $1 - \omega > 0$, load by the instrument density $\sqrt{E(y)}$; with probability ω , use a tagged rank-one projective instrument. Perform an informationally complete audit and include its average back-action. Later publicly identified trace-preserving positive maps may be noncommuting unitaries composed with that audit.

If physical data lifetimes have a uniform third moment, independent of these load innovations conditional on the entering control label, these experiments have global covers and finite occupation stability, without strict filter contraction. Their actual average memory rates for fixed control size are $J^{-2/d}$. Lower constants retain the actual load, recurrent-class and continuous-stratum masses. The same exact/effective boundary results allow common Cantor or countable report references with certified tails; arbitrary mutually singular adaptive path families are not thereby covered.

Proof. For (43), Bayes' rule gives $p_i = \pi_i g_i / \sum_j \pi_j g_j$ and the explicit inverse

$$g_i = \frac{(d+1)p_i/\pi_i}{\sum_j p_j/\pi_j}, \quad y_i = (g_i - 1)/\eta \quad (i \geq 1).$$

The compact cube is mapped smoothly with nonsingular derivative and bounded inverse derivative to its posterior image. Its report density is bounded above and below. Thus the actual first-posterior law has a bounded d -dimensional density, while indicator-audit tests separate the predictive means. Later stochastic maps are nonexpansive in ℓ^1 . Their entire simplex has global J -covers of radius $O(J^{-1/d})$. Report probabilities at the load are affine in the prior, hence continuous in total variation; later public innovations are independent of it. These are verifications from likelihoods, not an assumed Fisher geometry.

In the quantum case, $\int E(y) d\sigma = I$ and $\text{tr } E(y) = q$, so under I/q the continuous report is exactly uniform and its pre-audit state is $E(y)/q$. Its mass is $1 - \omega$, not one. The projective branch has its genuine pure-state atoms. With $F_{j,\pm} = (I \pm cH_j)/(2d)$ positive, these audit effects sum to I , and their linear expectation map is injective on the traceless difference space. Its least singular value is positive, as quantified in Appendix A. Hence actual executable audit means, not a postulated state parametrization, give dimension d . Positive trace-preserving maps are trace-norm nonexpansive on Hermitian differences by applying them to positive and negative parts. The compact density-matrix body has the required global trace-norm covers; load report laws are affine and continuous in the prior.

The known preparation allows the predictor coordinate to initialize exactly without erasing the exploration label. On subsequent nonexpansive calls G_t grows at most linearly in the data age; $\sum_{t \text{ data}} G_t^2 \leq C\tau^3$. The uniform conditional third moment thus proves (32) in every closed class.

One first load per cycle gives an actual positive submeasure in (28). Apply (35) and Theorem 5.2. If control labels split that submeasure among finitely many fibers, at least one carries positive mass; no mass is renormalized. All coordinates needed for the stationary update are either in the retained control state or are presently visible load/propagation/end types. This is a fixed-exploration verification, unless the extra conditions of Corollary 5.3 have also been established.

The tensor and kernel arguments use probability references and L^1 approximation, not Lebesgue density. Common Cantor or countable references work identically. Certified omitted report mass is an actual tail cell, not a conditionally normalized experiment. Rank tags may be added to these references. A measurable kernel-density construction and its precise scope are given in Appendix A. $\square$

Proposition 6.5 (Endogenous return and acquired geometry). *The matching geometric conclusion can be realized without an independent lifetime clock, in both raw classes.*

For the finite hidden state, first perform the positive load (43). Subsequently let action a have an alive substochastic matrix Q_a whose row sums lie in $[s_{\min}, s_{\max}]$. Its complementary end kernel sends the state to a fixed known terminal state. The visible report is alive or end; the actual indicator audit is not feedback. Assume (44). Under any fixed, counted exploration with a certified boundary matrix, its conditional checkpoint and online excess risks have matching order $J^{-2/d}$ at fixed control size. Lifetimes are state-dependent and may depend on the observed load through later feedback.

For a quantum version, first perform the continuous/rank-one load of Corollary 6.4, then use the noncommuting alive/jump instruments (42). Assume

$$(44) \quad 0 < s_{\min} \leq s_j \leq s_{\max} < 1, \quad s_{\max} \left(\frac{2s_{\max}}{s_{\min}} \right)^2 < 1.$$

An informationally complete audit is scored at the load and after each propagation, with its average back-action included. The same fixed-control matching rate holds with $d = q^2 - 1$ when the corresponding q -level diagonal instrument and independent noncommuting unitaries obey (44). No one-step contraction of the normalized alive update is required. For example $s_{\min} = 9/100$, $s_{\max} = 1/10$ satisfy the condition with normalized-update bound $20/9 > 1$.

Proof. The hidden load is fresh with the same bounded-density posterior law conditional on every incoming label. The alive update is $p \mapsto pQ_a/(pQ_a1)$ and the end update is constant. Since row sums are between $s_{\min}$ and $s_{\max}$, the alive denominator is bounded below and splitting the quotient difference gives the ℓ^1 Lipschitz bound $L = 2s_{\max}/s_{\min}$. Survival probability is at most $s_{\max}$ at every call for every chosen action. The flag law is affine in the belief and hence continuous in total variation, uniformly over actions. The indicator audit separates all belief coordinates, so here the compact simplex, together with the visible phase, realizes the full task-predictive quotient rather than an unverified covariance chart. The gain calculation below gives (32). There is one fresh full-dimensional load per cycle and mean length at most $\bar{\mu}$, so its actual occupation mass from (28) is at least $1/\bar{\mu}$. Feedback may alter later length bias but cannot erase this first-load submeasure. The global simplex cover and small-ball lower complete the hidden verification.

For the quantum construction, the fresh load and its actual continuous mass are unchanged; the later lifetime is state-dependent. Instrument report probabilities are affine in the input state and continuous in total variation. On the whole density-matrix body, the normalized alive map $X \mapsto KXK^*/\text{tr}(KXK^*)$ has trace-norm Lipschitz constant at most $L = 2s_{\max}/s_{\min}$. Indeed its numerator difference is bounded by $s_{\max}\|X - X'\|_1$, its denominator is at least $s_{\min}$, and splitting the quotient difference gives the factor two. An end jump has the constant normalized output $|0\rangle\langle 0|$ and hence Lipschitz constant zero. The audit's unreported average channel is nonexpansive. The probability of surviving t propagations, conditional on every previous history and action, is at most $s_{\max}^t$.

For the unit-error gain, $G_t \leq 1 + L + \dots + L^t \leq L^{t+1}/(L-1)$ because $L \geq 2$. Thus its actual squared occupation is bounded by a geometric series with ratio $s_{\max}L^2 < 1$. This verifies all-candidate stability on the full state domain although the normalized update need not contract. The global cover and first-load small-ball witness are unchanged. The mean length is bounded by the same alive hazard, so the first-load mass remains positive in every recurrent class.

For both constructions a primitive retained control architecture can keep its control label unchanged on the initial load and use the ending-row graph floors at propagation. The first propagation ends with probability at least $1 - s_{\max}$; the proof of Proposition 4.3 applies at this designated call because no earlier control-label change occurs. Predictor labels still update at load and are fully counted by the QJ lift. With finite visible phase types, all update and action rules remain stationary. This is a simultaneous raw verification of retained causal transfer and acquired geometry, not a physical reset imposed independently of state for the sake of a desired exponent. $\square$

The unbounded queue also supports a useful task-factor version: attach a fresh independent finite hidden load to each busy period and propagate it by the publicly selected stochastic actions. The queue drift supplies a conditional third moment, the hidden factor is nonexpansive, and its first-load law has mass at least $1/\bar{\mu}$. This gives the same fixed-control task-compression rate even when later queue actions depend on the load. It does not identify that finite hidden factor with the full predictive quotient of the unbounded workload; the workload report law remains in the original uncompressed experiment.

7. SHARP CONSEQUENCES AND THE LIMITS OF UNIFORMITY

7.1. Three distinctions forced by retained information.

Proposition 7.1 (Boundary scale and classwise physical time). *The recurrence term in Theorem 2.1 cannot in general be removed, even with deterministic bounded cycle length. The classwise ratios in (7) cannot in general be replaced by a pooled ratio. Compactness of the unrestricted program space alone does not imply continuity of its average losses.*

Proof. For $0 < \eta < 1/2$, take the retained boundary matrix $P_\eta = \begin{pmatrix} 1-\eta & \eta \\ \eta & 1-\eta \end{pmatrix}$, initial label zero, and reward indicator of label one, constant during a deterministic two-call cycle. Over n complete cycles the expected average is

$$(45) \quad \frac{1}{2} - \frac{1 - (1 - 2\eta)^n}{4\eta n}, \quad h(P_\eta) = \frac{1}{2\eta}.$$

Thus the error is of order $\min\{1, h(P_\eta)/n\}$. At $\eta = 0$ the initial label is absorbing and the average is zero, whereas it is $1/2$ for every $\eta > 0$. This proves both the necessity of a recurrence scale in a uniform bound and the possible failure of unrestricted continuity. The completely periodic swap matrix has a finite group inverse and is included in Theorem 3.2; period is not the obstruction.

For the second assertion, from an initial transient state enter either of two absorbing labels with probability $1/2$. Let their deterministic cycle lengths be two and four, and their per-call losses zero and one. The true average is $1/2$. The pooled ratio would be $(0 + 4)/(2 + 4) = 2/3$. The difference is physical class weighting, not a matter of notation.

For a perturbation counterpart, change one of the two rates η to $\eta + t$, $0 < t \leq \eta$. The stationary probability changes by $t/[2(2\eta + t)]$, whereas the one-cycle row total variation is t . Recurrence-sensitive amplification of order $h(P)t$ is therefore genuine. These finite matrices are witnesses for the general mechanism, not its proof. $\square$

A physical reset also does not make learning across cycles vacuous. For an unknown fixed Bernoulli bias, a retained finite-state rule may accumulate evidence across independent acquisitions.

Classical finite-memory hypothesis testing already treats this phenomenon [12]. In particular a sharpness example using a *fixed* hidden bit that is revealed on ordinary short cycles cannot be transferred unchanged from a compulsory-reset protocol: a retained observer can learn that bit and use it on later erased cycles. The next construction uses a fresh scored coordinate and explicitly specifies the common task wire.

7.2. A matched memory–calibration–defect slice. A *task-preserving* simulator below must pass through the environment-owned audit coordinate unchanged. It may transform visible reports and maintain private labels, but may not synthesize a replacement ground-truth audit. The target’s load report must be completed before its first scored audit. This is the common-task causal interface being compared, not unrestricted Blackwell simulation with freely replaceable audit output.

Theorem 7.2 (An attained nonreset joint resource law). *Let a fresh vector V at every physical preparation have a compact law π with $q_M(\pi) \asymp M^{-2/d}$. Independently draw a fresh hidden fair bit B and a lifetime L , with $L = 1$ except on a marked rare event of probability $e \in (0, 1/2]$, on which $L = \ell \geq 1$. Preparation is charged and has zero score. At load the vector is visible. In the target, the bit is also visible; in the source it is replaced by a tagged erasure on rare cycles. Later visible holding reports contain neither a new vector nor information about the erased bit. At each of L scored data calls the actual audit target is (V, u, B) , where $u \in \{-\delta, \delta\}$ is one fixed inaccessible calibration parameter and $0 \leq \delta \leq 1$. Audits are not feedback. The observer is not overwritten at boundaries.*

For task-preserving simulators the conditional completed-cycle deficiency of source to target is exactly

$$(46) \quad \varepsilon = e/2.$$

Put $\mu = 1 + \mathbb{E}L$ and $w = \mathbb{E}L/\mu$. Over all stationary retained M -state observers and both fixed calibration alternatives, the unnormalized squared risk satisfies

$$(47) \quad wq_M(\pi) + w\delta^2 + \frac{e\ell}{4\mu} \leq \mathcal{R}_{\text{av}}(M, \delta, e),$$

$$\mathcal{R}_{\text{av}}(M, \delta, e) \leq wq_J(\pi) + w\delta^2 + \frac{e\ell}{4\mu}, \quad J = \lfloor M/3 \rfloor, \quad M \geq 3.$$

The upper program uses at most $3J$ labels and has $h(P) \leq 2$; its preparation label is one of those existing labels because preparation is unscored and load/hold types are visible. The lower allows arbitrary retained controllers and fresh randomization, not merely the upper architecture.

For $0 < p \leq 1$, choose $\ell = \lceil e^{-1/(1+p)} \rceil$. The family has a uniform $1 + p$ cycle moment, and

$$(48) \quad \mathcal{R}_{\text{av}}(M, \delta, \varepsilon) \asymp M^{-2/d} + \delta^2 + \varepsilon^{p/(1+p)} \quad (M \geq 3).$$

All constants can be uniform in e, δ, M with the displayed preparation law and p fixed. Dividing the task by the fixed bound $\text{diam}(\text{supp } \pi)^2 + 5$ gives a loss in $[0, 1]$. The finite-acquisition comparison on a certified recurrence class is the signed error of Theorem 2.1, not an additional positive learning penalty.

Proof. Conditional on any incoming observer or simulator label, the new bit is fair and independent of all past cycles and the vector. On a rare source load, no permitted information before the first audit reveals it. A guessed target bit agrees with the unaltered audit bit with probability at most $1/2$. In the target this agreement is certain. The measurable event “rare and load bit equals first audit bit” therefore gives total variation at least $e/2$ for the complete common-task law. Guessing a fresh fair bit on rare cycles and passing the true audit wire unchanged attains that distance; on ordinary cycles simply pass through the reported bit. Repeating the guess consistently on that cycle uses a finite simulator state, which is counted in a composed implementation. The audit

deadline and identity wire are indispensable: if an independent artificial audit could be substituted, this would not be the same deficiency problem.

For any observer, its at most M vector responses have pointwise squared error at least their nearest-center distortion. The vector is fresh and independent of lifetime and all incoming labels, so the acquired vector submeasure is $w\pi$. For the calibration lower, average the two fixed alternatives only as a minimax lower device. Their visible laws are identical, hence $\frac{1}{2}[(a - \delta)^2 + (a + \delta)^2] = a^2 + \delta^2 \geq \delta^2$ at every scored call. The parameter is not redrawn during execution. On each rare cycle the fresh bit remains independent of the entire visible history throughout its holds, and any response has mean squared error at least $1/4$. Summing these three coordinate lower bounds gives the first line of (47). Retained memory cannot invalidate freshness of the current bit; that is exactly the difference from a fixed hidden-bit example.

For the upper, use J vector centers crossed with bit outputs $0, 1, 1/2$, and calibration output zero. At load choose the nearest vector center and the reported bit, or $1/2$ after erasure; hold that label thereafter. The total count is $3J \leq M$, and the already existing label at preparation incurs no score. Its next-cycle boundary label distribution is independent of its input label because every fresh load replaces it voluntarily. Thus $P = \Pi_P$ and $h(P) = \|I - P\|_\infty \leq 2$. This proves the upper line.

Finally $\ell \leq 2e^{-1/(1+p)}$ and $\ell + 1 \leq 3e^{-1/(1+p)}$, so $\mathbb{E}\tau^{1+p} \leq 2^{1+p} + 3^{1+p}$. Moreover μ is bounded above and below uniformly, $w \geq 1/2$, and $e\ell \asymp e^{p/(1+p)}$. The comparison of q_J and q_M gives (48). Readouts may be restricted to the convex task ranges; the vector error is at most its squared diameter, calibration error at most $4\delta^2 \leq 4$, and bit error at most one. This proves the normalization and its uniformity for all admissible programs. $\square$

Proposition 7.3 (Return-moment sharpness without memory erasure). *Uniformly over experiments with a bounded $1 + p$ cycle moment, neither the N^{-p} nor the $(1 - \beta)^p$ order in Theorem 2.1 can be improved. These are class-sharp orders, not lower bounds for every fixed experiment.*

Proof. One retained label suffices. Let a cycle have one unscored preparation and either one zero-loss data call or ℓ unit-loss data calls; the long case has probability $e = \ell^{-(1+p)}$. The cycle moment is uniformly bounded and its long-run loss is $g = e\ell/[2 + e(\ell - 1)] \asymp \ell^{-p}$. Before N calls there are at most N possible cycle starts. The probability of any rare start is at most Ne , so $J_N \leq Ne$. Choose $N = \lfloor c\ell \rfloor$ for a sufficiently small fixed $c > 0$. Then $g - J_N \geq c'\ell^{-p} \asymp N^{-p}$. The lower uses an event probability, not a count of rewards per possible start.

For normalized discount, the probability of a rare start before call t is at most te , whence $J_\beta \leq e/(1 - \beta)$. Choose $(1 - \beta)^{-1} = c\ell$ with the same sufficiently small c . This again gives $g - J_\beta \geq c'\ell^{-p} \asymp (1 - \beta)^p$. There is no observer state to erase in this example, so it belongs to the retained class with $H = 0$. The sharpness of recurrence-sensitive perturbation and of the tail exponent are separate, as the two examples show. $\square$

8. COMPARISON, SCOPE, AND CONSEQUENCES

The theorem is a nonreset observer theorem, not a theorem without physical regeneration. At a boundary the physical innovations restart conditional on one fixed parameter; the observer's label survives and can carry evidence and control information. Multiple closed observer classes are handled by their separate physical-time ratios. No filter contraction or irreducibility of every boundary matrix is assumed. Uniformity instead has the exact cost-universal boundary criterion of Theorem 3.2, together with the raw physical-tail condition.

The established neighboring results answer different questions. Table 1 summarizes the distinctions; “fixed memory” means an externally specified persistent-state cardinality, not memory chosen after an accuracy target. Recent preprints are identified as such and no priority conclusion is inferred from their omission of our notation.

Result	Strategy and objective	Distinction from the present theorem
Chatterjee–Saona–Ziliotto [4]	Finite-memory approximate optimality for long-run POMDP value	Memory may depend on accuracy; not our fixed- M robust cycle certificate
Venel–Ziliotto [5]	Strong uniform value and pure approximately optimal strategies	Broader strategy comparison; not a recurrence sublevel of one stored program
Borkar [6]; Yu–Bertsekas [7]	Average-cost equations; near-optimal finite controllers	Complementary dynamic programming and approximation assumptions
Yüksel [8]	History-space existence, ergodicity and approximations	Does not identify a free belief or history state with fixed observer memory
Avrachenkov et al. [9, 10]	Controlled restart; intermittent observation and average reward	Restart choice and elapsed-information age differ from physical renewal with retained labels
Asadi–Chatterjee–Lurie [11]	Revealing-POMDP value approximation, EXPTIME-completeness	The effective input and strategy class differ; our enumeration is not an efficient solver for that class
Present result	One robust retained M -state program; average, raw-call and discount objectives	Conditional cycle laws, boundary recurrence, classwise time, same- M certified implementation

TABLE 1. Assumptions and conclusions, rather than shared terminology.

The group inverse, hitting identities, renewal reward and directed forest expansion are standard tools [1, 2, 3]. Convexity is not used to make a nonlinear feedback program space convex. Product response continuity supplies a legitimate compact topology for one reused Borel row; recurrence sublevels then prevent loss of average continuity when a closed class appears. The finite-model identity is the finite-intersection consequence in this one common space. It does not say that each model may select its own controller.

The synthesis theorem is global at a fixed label count on a support-stable architecture. Its lower certificate covers every Borel program in that architecture, and its upper table uses the original experiment and the same labels. Uniformly computable marked response errors, return tails, model nets and integrations are nevertheless *inputs*. The queue and instrument verifications derive such inputs from primitive data in explicit classes. Enumeration (26) proves effectivity, not a polynomial upper bound, and not a complexity classification for unrestricted partially observed control. This separation is especially important in light of the neighboring revealing-POMDP complexity result [11].

For predictive geometry, (28) is the measure that is actually acquired in physical time. A local small-ball bound can give a lower rate only with its actual mass. The recursive upper also needs a global candidate-valid cover and an occupation-weighted stability bound. These sufficient conditions permit noncontractive propagation, singular strata and long holding times. They are not claimed to be a necessary-and-sufficient classification of recursively attainable checkpoint geometry. The optimized inequalities (36)–(37) state exactly where control enters and where an additional common attaining exploration is needed.

The resulting general resource description has two levels. On a declared recurrence class,

$$\Phi_{\mathfrak{C}}(M, H) = \min_{\gamma \in \Gamma_{M, H}^{\Theta}} \sup_{\theta} e_1 \Pi_{S_{\theta}^{\gamma}} (D_{\theta}^{\gamma})^{-1} r_{\theta}^{\gamma}$$

is attained, has the finite-model characterization, and differs from the same-budget finite-acquisition risk by at most b_N . On the classes of Theorem 4.4 it has arbitrarily tight implementable and information-theoretic certificates. Geometry supplies explicit matching rates under Corollary 5.3;

Theorem 7.2 realizes memory, calibration and an attained task-preserving defect in one task. These statements do not assert a universal exponent in the ambient parameter dimension.

Remaining mathematical questions are genuine. The fixed-memory unrestricted value can be exhausted for finite model families, but uniform rates and attainment need not survive $H \rightarrow \infty$. General nondominated adaptive response families, a complete intrinsic recursive-geometry criterion, efficient synthesis, and optimal joint lower bounds for program description, workspace, arithmetic precision, physical time and minimal simulator state are not established here. Neither TV-to-LDP equivalence nor an unbounded-operator closure follows from our finite-dimensional comparisons. These limitations do not alter the experiment-to-prediction-to-resource question; they delimit the proved transfer precisely.

APPENDIX A. MEASURABILITY, OCCUPATION TIMING, AND EXPLICIT CONSTANTS

A.1. Kernel versions.

Lemma A.1 (A measurable dominated kernel density). *Let U, Y be standard Borel and σ a probability on Y . For a measurable finite kernel $u \mapsto K_u \ll \sigma$, there is a jointly measurable density representing each K_u up to its own σ -null set.*

Proof. Choose a countable algebra generating the Borel field of Y and let $\mathcal{P}_n$ be the finite partition generated by its first n sets. On a positive-mass cell C set $f_n(u, y) = K_u(C)/\sigma(C)$, and set it to zero on null cells. Each f_n is jointly measurable. For fixed u , these are the conditional expectations of any Radon–Nikodym density of K_u . The martingale convergence theorem gives its almost-sure limit and L^1 convergence. The joint liminf, set to zero wherever infinite, is a jointly measurable version. The exceptional set may depend on u ; no common null set over all parameters is claimed. Standard Borel disintegration, as developed in [13, 14], then gives sequential finite-word densities by applying this construction to successive conditional hidden-history kernels. Conditioning on an entire hidden path as if it were independent of earlier reports is unnecessary. This proof explicitly supplies the generating-partition hypotheses rather than relying on an unspecified kernel-version assertion. $\square$

A.2. The queue moments. For the biased walk with up probability $p < 1/2$, down probability $q = 1 - p$, put $b = q - p$. Starting at workload w , the hitting time T_w of zero has mean $m_w = w/b$. Its second moment is

$$(49) \quad \mathbb{E}T_w^2 = \frac{w^2}{b^2} + \frac{4pqw}{b^3}.$$

Indeed the equations with zero boundary values are

$$m_w = 1 + pm_{w+1} + qm_{w-1}, \quad s_w = 1 + 2(pm_{w+1} + qm_{w-1}) + ps_{w+1} + qs_{w-1}.$$

Substitution gives the displayed polynomial solutions. Stopping on a finite interval and then using the exponential Lyapunov bound proves their integrability and identifies them with the hitting-time moments, rather than merely formal solutions of a recurrence. At $p = 1/4$, $w = 1$, they give $m_1 = 2$, $s_1 = 10$, and the preparation-adjusted bounds used in (41).

A.3. Load timing and nonexpansive propagation. For the iid-lifetime load models of Corollary 6.4, assume $\tau = L + 1$ has mean μ and the first data law is independent of the lifetime. Its average occupation weight is $1/\mu$. At a finite external horizon $N \geq 2$, renewal epochs $S_k \leq N - 2$ each give one first-data report. If $K_x = \min\{k : S_k > x\}$, then $\mathbb{E}K_x \geq x/\mu$ by the stopped sum identity and $S_{K_x} > x$, while there is always the initial epoch. Hence the weight of that first-data law is at least

$$(50) \quad \frac{1}{N} \max\{1, (N - 2)/\mu\} \geq \frac{1}{2\mu}.$$

For $N = 2, 3$ use the first term and $\mu \geq 2$; for $N \geq 4$ use the second. At normalized discount the same weight is exactly

$$(51) \quad (1 - \beta) \sum_{k \geq 0} \mathbb{E} \beta^{S_k+1} = \frac{\beta(1 - \beta)}{1 - \mathbb{E} \beta^\tau} \geq \frac{\beta}{\mu},$$

because $1 - \beta^\tau \leq \tau(1 - \beta)$. Thus it is at least $1/(2\mu)$ for $\beta \geq 1/2$. A continuous quantum branch multiplies each weight by its actual factor $1 - \omega$.

For the corresponding nonexpansive recursive upper, error at data age t is at most $Cr_J t$. For finite N , the expected total of its squared gains is bounded by summing over at most N possible renewal epochs, each contributing at most $C^2 \mathbb{E} \tau^3$. Dividing by N gives a uniform constant; the positive Abel mixture gives the discount version. For retained exploration with label-dependent lifetimes the main average theorem uses the class-weighted law (28); the simple exact first-weight formulas above are stated only for the iid-lifetime subclass. They are not silently applied to arbitrary Markov-renewal lifetimes.

A.4. Informational completeness and physical back-action. For $d = q^2 - 1$ linearly independent traceless Hermitian H_j , the effects $F_{j,\pm} = (I \pm cH_j)/(2d)$ satisfy $\sum_{j,\pm} F_{j,\pm} = I$. On the traceless Hermitian space with Hilbert–Schmidt norm, let $s_0 > 0$ be the least singular value of $X \mapsto (\text{tr } H_j X)_j$. The audit map $f(X) = (\text{tr } F_{j,\pm} X)_{j,\pm}$ therefore obeys

$$(52) \quad \|f(X)\|_2 \geq \frac{cs_0}{\sqrt{2}d} \|X\|_2.$$

All norms on this finite-dimensional difference space are equivalent, giving the inverse Lipschitz constant required in Corollary 6.4. The performed audit changes the subsequent physical state by its completely positive average channel; it is not a simultaneous readout of incompatible unperformed measurements.

A.5. Scope of the inherited reset comparison. The complete prior regenerative article remains available as a separate companion. Its compulsory observer overwrite makes its cycles iid and permits fixed hidden-bit alternatives to be used afresh as a minimax comparison each cycle. Theorem 7.2 changes that sharpness construction for the retained-state interface. The original fixed-bit result is not a premise of the new lower bound. Likewise, the previous scalar renewal ratio is a one-boundary-state case; it is not the multichain ratio. The preceding timing and kernel details expand the same elementary facts without deleting or changing their historical statements.

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